

LCM and HCF

1. Which one is the largest fraction among $\frac{5}{7}$, $\frac{7}{9}$ and $\frac{11}{13}$?
(A) $\frac{5}{7}$ (B) $\frac{7}{9}$
(C) $\frac{11}{13}$ (D) All are equal
2. What is the Highest Common Factor of 24, 40 and 120?
(A) 8 (B) 4
(C) 12 (D) 40
3. What is the LCM of $\frac{18}{5}$ and $\frac{20}{9}$?
(A) 60 (B) 12
(C) 30 (D) 180
4. Which fraction among $\frac{3}{7}$, $\frac{4}{11}$ and $\frac{5}{8}$ is the smallest?
(A) $\frac{3}{7}$ (B) $\frac{4}{11}$
(C) $\frac{5}{8}$ (D) All are equal
5. What is the Highest Common Factor of $\frac{3}{4}$ and $\frac{12}{13}$?
(A) $\frac{3}{13}$ (B) $\frac{3}{52}$
(C) $\frac{3}{26}$ (D) $\frac{3}{41}$
6. Which fraction among $\frac{3}{11}$, $\frac{4}{7}$ and $\frac{5}{8}$ is the smallest?
(A) $\frac{3}{11}$ (B) $\frac{4}{7}$
(C) $\frac{5}{8}$ (D) All are equal
7. Which of the following is true?
(A) $\frac{6}{7} > \frac{4}{5} > \frac{3}{4}$ (B) $\frac{4}{5} > \frac{6}{7} > \frac{3}{4}$
(C) $\frac{6}{7} > \frac{3}{4} > \frac{4}{5}$ (D) $\frac{3}{4} > \frac{4}{5} > \frac{6}{7}$
8. What is the least common multiple of 24, 48 and 72?
(A) 72 (B) 144
(C) 288 (D) 216
9. What is the LCM of 64 and 56?
(A) 448 (B) 488
(C) 484 (D) 408
10. What is the HCF of 6345 and 2160?
(A) 45 (B) 135
(C) 270 (D) 15
11. What is the LCM of 120 and 450?
(A) 2400 (B) 1800
(C) 3600 (D) 4800
12. What is the HCF of $2^3 \times 3^4$ and $2^5 \times 3^2$?
(A) $2^5 \times 3^3$ (B) $2^3 \times 3^4$
(C) $2^3 \times 3^2$ (D) $2^5 \times 3^4$
13. A is The highest common factor of $\frac{3}{4}$ and $\frac{9}{16}$, and least common multiple of $\frac{16}{5}$ and $\frac{4}{25}$ is B, then what is the value of A + B?
(A) $\frac{250}{81}$ (B) $\frac{70}{23}$
(C) $\frac{271}{80}$ (D) $\frac{260}{71}$
14. What is the maximum common factor (HCF) of $\frac{7}{16}$, $\frac{21}{32}$ and $\frac{49}{8}$?
(A) $\frac{7}{64}$ (B) $\frac{147}{32}$
(C) $\frac{147}{8}$ (D) $\frac{7}{32}$
15. What is the Least Common Multiple of all even numbers between 5 and 13?
(A) 120 (B) 90
(C) 180 (D) 60
16. If the Least Common Multiple of 56, 57 and 58 is K, then what will be the Least Common Multiple of 56, 57, 58 and 59?
(A) 177 K (B) 59 K
(C) 56 K (D) 57 K
17. If Least Common Multiple of 23 and 24 is A and Highest Common Factor of 23 and 24 is B, then what is the value of A + B?
(A) 451 (B) 551
(C) 553 (D) 452

18. What is the Highest Common Factor of 42, 168 and 210?
(A) 14 (B) 21
(C) 42 (D) 7
19. The product of two numbers is 6845, if the HCF of the number is 37, then the greater number is:
20. If $x \times y$ denotes HCF of x and y and $x @ y$ denotes LCM of x and y , then the value of $(72 \times 84) @ 144$ is:
(A) 144 (B) 504
(C) 210 (D) 420

LCM and HCF

Solution

1. Answer: (C)

So, $11/13$ is the largest fraction

$$\frac{5}{7} = 0.714, \frac{7}{9} = 0.777, \frac{11}{13} = 0.846$$

Also we can solve it by equating the bases of these fractions.

LCM of 7, 9 & 13 = 819

$$\frac{5}{7} = \frac{585}{819}, \frac{7}{9} = \frac{637}{819}, \frac{11}{13} = \frac{693}{819}$$

Here also we can see $693/819$ i.e. $11/13$ is the largest fraction

2. Answer: (A)

HCF of 24, 40 & 120

$$24 = 2 \times 2 \times 2 \times 3 = 2^3 \times 3^1$$

$$40 = 2 \times 2 \times 2 \times 5 = 2^3 \times 5^1$$

$$120 = 2 \times 2 \times 2 \times 3 \times 5 = 2^3 \times 3^1 \times 5^1$$

So, HCF = $2^3 = 8$ (since that is common in all the numbers)

3. Answer: (D)

For finding the LCM of 2 fractions.

LCM = LCM of Numerators/HCF of Denominators

LCM = LCM of 18 & 20/HCF of 5 & 9 = $180/1 = 180$

4. Answer: (B)

$$3/7 = 0.4284, 4/11 = 0.3636, 5/8 = 0.625$$

So smallest is $4/11$

It can also be solved by making the denominators equal

LCM of 7, 11 & 8 = 616

$$\frac{264}{616} = \frac{3}{7}, \frac{224}{616} = \frac{4}{11}, \frac{385}{616} = \frac{5}{8}$$

When, bases are equal, we can clearly see.

$224/616$ is the smallest i.e. $4/11$

5. Answer: (B)

For fraction,

H.C.F = H.C.F of numerator/L.C.M of denominator

Numerators are 3 and 12

H.C.F of 3 and 12 = 3

Denominators are 4 and 13

L.C.M of 4 and 13 = 52

H.C.F = $3/52$

6. Answer: (A)

$$3/11 = 0.27$$

$$4/7 = 0.57$$

$$5/8 = 0.625$$

We can observe that $3/11$ has the minimum value.

$\therefore 3/11$ is the smallest fraction among all

7. Answer: (A)

$$6/7 = 0.85$$

$$4/5 = 0.8$$

$$3/4 = 0.75$$

Here, $0.85 > 0.8 > 0.75$

Hence, $6/7 > 4/5 > 3/4$ is true.

8. Answer: (B)

Listing the prime factor of each number

$$24 = 2 \times 2 \times 2 \times 3$$

$$48 = 2 \times 2 \times 2 \times 2 \times 3$$

$$72 = 2 \times 2 \times 2 \times 3 \times 3$$

Multiply each factor the greatest number of times it occurs in any of the numbers.

48 has 4 2s, 72 has 2 3s. This gives us 144, the smallest number that can be divided evenly by 24, 48 and 72

9. Answer: (A)

2	64, 56
2	32, 28
2	16, 14
	8, 7

Required LCM = $2 \times 2 \times 2 \times 2 \times 8 \times 7 = 448$

10. Answer: (B)

HCF of 6345 and 2160

Factors of 6345 = $5 \times 3 \times 3 \times 3 \times 47$

Factors of 2160 = $5 \times 3 \times 3 \times 3 \times 2 \times 2 \times 2 \times 2$

So, common factors = $5 \times 3 \times 3 \times 3$
= 135

11. **Answer: (B)**

5	120,450
2	24,90
3	12,45
3	4,15
4	4,5
5	1,5
	1,1

$$\text{LCM} = 5 \times 2 \times 3 \times 3 \times 4 \times 5 = 1800$$

12. **Answer: (C)**

$$A = 2^3 \times 3^4$$

$$B = 2^5 \times 3^2$$

$$\text{HCF of } (A, B) = 2^3 \times 3^2$$

13. **Answer: (C)**

$$A = \frac{\text{HCF}(3,9)}{\text{LCM}(4,16)}$$

$$= \frac{3}{16}$$

$$B = \frac{\text{HCF}(16,4)}{\text{LCM}(5,25)}$$

$$= \frac{16}{5}$$

$$\text{So, } A + B = \frac{3}{16} + \frac{16}{5}$$

$$= \frac{15+256}{80}$$

$$= \frac{271}{80}$$

14. **Answer: (D)**

$$\text{HCF of } \left(\frac{p}{q}\right) = \frac{\text{HFC of Numerator}}{\text{LCM of Denominator}}$$

$$\text{HFC} = \frac{\text{HCF}(7,21,49)}{\text{LCM}(16,32,8)}$$

$$\text{HCF} = \frac{7}{32}$$

15. **Answer: (A)**

Even number between 5 and 13 $\therefore$ 6, 8, 10, 12

$$\text{LCM}(6, 8, 10, 12) = 120$$

16. **Answer: (B)**

$$\text{LCM}(56, 57, 58) = K$$

$$\text{LCM}(56, 57, 58, 59)$$

$$= \frac{56, 57, 58}{K} \frac{59}{59}$$

$$\text{LCM}(56, 57, 58, 59) = 59K$$

17. **Answer: (C)**

$$\text{LCM}(23, 24) = 552$$

$$A = 552$$

$$\text{HCF}(23, 24) = 1$$

$$B = 1$$

$$\text{A.T.Q}$$

$$= A + B$$

$$= 552 + 1$$

$$= 553$$

18. **Answer: (C)**

$$\text{H.C.F}(42, 168, 210)$$

$$42 = 21 \times 2$$

$$168 = 21 \times 2 \times 2 \times 2$$

$$210 = 21 \times 2 \times 5$$

$$\text{HCF}(42, 168, 210) = 42$$

19. **Answer: (D)**

Let the number are x and y

$$37xy = 6845$$

$$xy = 185$$

$$xy = 37 \times 5$$

$$\text{1st Number} = 185$$

$$\text{2nd Number} = 37$$

20. **Answer: (A)**

$$\text{A.T.Q}$$

$$(72 \times 84) @ 144$$

$$\text{HCF of } 72 \text{ and } 84 = 12$$

$$12 @ 144$$

$$\text{LCM of } 12 \text{ and } 144 = 144$$

$$12 @ 144 = 144$$